Final

DIPLOMA CURRICULUM OF COMPUTER ENGINEERING AND IOT (THIRD YEAR) (5th Semester)

(To be implemented from 2026-27)

Prepared by;

National Institute of Technical Teachers' Training & Research Kolkata
Block – FC, Sector – III, Salt Lake City, Kolkata – 700106

Vetted by:
Domain experts from Polytechnics of Odisha

State Council for Technical Education & Vocational Training
Near Raj Bhawan, Unit-VIII, Bhubaneswar, Odisha

Table of Contents

PROGRAMME TITLE: COMPUTER ENGINEERING AND IOT

SEMESTER – V

SL. No	Category of Course	Code No	Course Title	Study Scheme: Pre-requisite	Contact Hours/ week: L	T	P	Evaluation Scheme: Theory: End Exam	Theory: Progressive Assessment	Practical: End Exam	Practical: Progressive Assessment	Total Marks	Credits
1	Program Core	CEIPC 301 TH1	Programming with Python		3	0	0	70	30	-	-	100	3
2		CEIPC 303 TH2	Web Technologies		3	0	0	70	30	-	-	100	3
3		CEIPC 305 PR1	Programming with Python Lab		0	0	4	-	-	15	35	50	2
4		CEIPC 307 PR2	Web Technologies Lab		0	0	4	-	-	15	35	50	2
5	Program Elective	CEIPE 301 Any One TH3	(A)Programming with Java (B)IoT using RASBERRY PI (C)Industry IoT and IoT Standards		3	0	0	70	30	-	-	100	3
6		CEIPE 303 Any One TH4	(A)Data Science and Data Management for IoT (B)IoT Communication Technology (C)Drone Technology and WSN		3	0	0	70	30	-	-	100	3
7		CEIPE 305 Any One PR3	(A)Programming with Java Lab (B)IoT using RASBERRY PI Lab (C)Industry IoT and IoT Standards Lab		0	0	4	-	-	15	35	50	2
8	Open Elective	OE 301 (Any one) TH5	(A)Universal Human Values (B) Leadership and Management Skills (C)Professional Skills		3	0	0	70	30	-	-	100	3
9	Summer Internship	SI 301	Summer Internship-II*		0	0	0	-	-	15	35	50	2
10	Major Project	PR 301 PR4	Major Project		0	0	4	-	-	15	35	50	2
TOTAL					**15**	**0**	**16**	**350**	**150**	**75**	**175**	**750**	**25**

***3-4-week internship after 4th Semester**

***The best of 2 IA conducted in a subject out of 20 marks to be considered. Assignment/ quiz etc. of 10 marks to be treated as part of IA. Besides this, Monthly Test to be conducted for each subject. Sessional Marks shall be total of the performance of individual different jobs/ experiments in a subject throughout the semester. Club/Innovation/ Idea Tinkering Activities etc. shall be encouraged to be performed by students beyond the above stipulated hours.**

SEMESTER - V COURSES

TH1- PROGRAMMING WITH PYTHON

L	T	P		
3	0	0	**Total Marks: 100**	**Course Code: CEIPC 301**
Total Contact Hours				**Theory Assessment**
Theory : 45 Hrs.				End Term Exam 70
				Progressive Assessment 30
Pre-Requisite : Nil				
Credit 3				Category of Course : PC

RATIONALE:

Python is a versatile and beginner-friendly scripting language, renowned for its simplicity and readability. It enables rapid development of applications in fields like web development, data analysis, automation, and artificial intelligence. Learning Python equips students with a powerful toolset to solve complex problems and adapt to evolving technology demands.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Define Python's core syntax, data types, and key concepts of object-oriented programming.
- Explain how control structures and data structures function in Python.
- Implement Python programs using file handling, modules, and libraries like NumPy, Pandas, and Matplotlib for data analysis and visualization.
- Analyze Python scripts to identify and resolve logical or syntactic errors and optimize code using advanced techniques like recursion and lambda functions.
- Develop a real-world mini-project by integrating Python concepts such as OOP, libraries, and automation tools for practical problem-solving.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to Python:** Overview of Python: Features and Applications Setting Up the Python Environment (Python Installation, IDEs), Python Syntax: Variables, Data Types, and Operators Writing, Executing, and Debugging Python Scripts	7
II	**Control Structures and Functions:** Conditional Statements: if, else, elif Loops: for, while, and Nested Loops Functions: Defining, Calling, and Scope of Variables Introduction to Lambda Functions and Recursion	7
III	**Data Structures in Python:** Lists, Tuples, Sets, and Dictionaries: Operations and Applications, List Comprehensions and Dictionary Comprehensions Working with Strings: Methods and Manipulation,	7

	Introduction to Python's Collections Module	

IV	**File Handling and Modules:** File Operations, Reading, Writing, and Appending Files Working with CSV and JSON Files, Python Modules: Built-In Modules (e.g., math, os, datetime) Creating and Using Custom Modules	8
V	**Object-Oriented Programming (OOP) in Python:** Understanding Classes and Objects Concepts of Encapsulation, Inheritance, and Polymorphism, Working with Magic Methods and Operator Overloading, Exception Handling in Python	8
VI	**Advanced Python and Applications:** Introduction to Libraries: NumPy, Pandas, Matplotlib Basics of Web Scraping: Using requests and BeautifulSoup, Scripting for Automation: Working with os and shutil Modules, Mini-Project: Developing a Python Script for a Real-World Problem	8

REFERENCES:

1.	John M. Zelle, Python Programming: An Introduction to Computer Science, 2nd Edition, Franklin, Beedle & Associates Inc., Portland, 2010.
2.	Al Sweigart, Automate the Boring Stuff with Python, 2nd Edition, No Starch Press, San Francisco, 2019.
3.	Reema Thareja, Python Programming Using Problem Solving Approach, Oxford University Press, New Delhi, 2017.
4.	Sheetal Taneja and Naveen Kumar, Python Programming: A Modular Approach with Graphics, Database, Mobile, and Web Applications, Pearson India, New Delhi, 2018.
5.	R. Nageswara Rao, Core Python Programming, Dreamtech Press, New Delhi, 2018.
6.	A.K. Sharma, Python for Beginners: With Hands-On Examples, BPB Publications, New Delhi, 2020.

TH2- WEB TECHNOLOGIES

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="7">Total Marks: 100</td><td rowspan="2" colspan="2">Course Code: CEIPC 303</td></tr>
<tr><td>3</td><td>0</td><td>0</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td colspan="2">Theory Assessment</td></tr>
<tr><td colspan="2">Theory</td><td>: 45 Hrs</td><td>End Term Exam</td><td>70</td></tr>
<tr><td colspan="3"></td><td>Progressive Assessment</td><td>30</td></tr>
<tr><td colspan="2">Pre-Requisite</td><td>: Nil</td><td colspan="2"></td></tr>
<tr><td colspan="2">Credit</td><td>3</td><td>Category of Course</td><td>: PC</td></tr>
</table>

RATIONALE:

This course provides a comprehensive understanding of web technologies, covering both client-side and server-side development. Students will learn web architecture, scripting languages like JavaScript and PHP, and database integration to build dynamic and interactive web applications. By the end of the course, they will be equipped with the skills to develop secure, scalable, and efficient web-based systems.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Identify key web development protocols, technologies, and architectures.
- Explain the role of client-side and server-side scripting in dynamic web applications.
- Develop interactive web pages using HTML, CSS, JavaScript, and PHP.
- Compare different web application architectures and evaluate their scalability and security.
- Develop a functional, database-driven web application using modern web development tools.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to www:** Protocols and programs, secure connections, application and development tools, the web browser, what is server, setting up UNIX and LINUX web servers, Logging users, dynamic IP Web Design: Website design principles, planning the site and navigation	8
II	**Web Systems Architecture:** Architecture of Web based systems- client/server (2-tier) architecture, 3-Tier architecture, Building blocks of fast and scalable data access Concepts - Caches-Proxies- Indexes-Load Balancers- Queues, Web Application architecture (WAA)	8
III	**JavaScript:** Client-side scripting, what is JavaScript, simple JavaScript, variables, functions, conditions, loops and repetition.	8

IV	**Advance scripting:** JavaScript and objects, JavaScript own objects, DOM and web browser environments, forms and validations, DHTML: Combining HTML, CSS and JavaScript, events and buttons, controlling your browser, Ajax: Introduction advantages & disadvantages, Ajax based web application, alternatives of Ajax, XML, XSL and XSLT: Introduction to XML, uses of XML, simple XML, XML key components, DTD and Schemas, XML with application, XSL and XSLT. Introduction to Web Services	8
V	**PHP:** Server-side scripting, Arrays, function and forms, advance PHP Databases - Basic command with PHP examples, Connection to server, creating database, selecting a database, listing database, listing table names creating a table, inserting data, altering tables, queries, deleting Database, deleting data and tables, PHP my admin and database bugs.	8
VI	**Introduction to Modern Frameworks:** React, Angular, Node JS, Bootstrap, Tailwind CSS.	5

REFERENCES:

1.	"Web Technologies--A Computer Science Perspective", Jeffrey C.Jackson
2.	"Internet & World Wide Web How to Program", Deitel, Deitel, Goldberg, Pearson Education
3.	"Web programming- Building Internet Application", Chris Bales
4.	Web Applications: Concepts and Real-World Design, Knuckles.

PR1- PROGRAMMING WITH PYTHON LAB

L	T	P		
0	0	4	Total Marks: 50	**Course Code: CEIPC 305**
Total Contact Hours				**Practical Assessment**
Practical : 60 Hrs				End Exam 15
				Progressive Assessment 35
Pre-Requisite : Nil				
Credit 2				Category of Course : PC

RATIONALE:

Python is a versatile and beginner-friendly scripting language, renowned for its simplicity and readability. It enables rapid development of applications in fields like web development, data analysis, automation, and artificial intelligence. Learning Python equips students with a powerful toolset to solve complex problems and adapt to evolving technology demands.

Learning Objectives:

After completion of the course, the students will be able to:

- Describe Python's features, applications, and fundamental concepts such as control structures, functions, and recursion.
- Develop Python programs using built-in data structures, string manipulations, file handling, and modules for practical problem-solving.
- Examine and implement object-oriented programming concepts, including classes, objects, inheritance, and exception handling, to structure programs effectively.
- Evaluate advanced Python libraries such as NumPy, Pandas, and Matplotlib to process, analyze, and visualize data for specific applications.
- Develop a mini-project to integrate Python programming concepts, demonstrating real-world applications and problem-solving capabilities.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to Python:** • Install Python and set up an IDE (e.g., PyCharm, VS Code, Jupyter, Spyder) • Write simple Python scripts to demonstrate variable declarations • Data types, and operators • Debug Python scripts to identify and fix errors.	10

II	**Control Structures and Functions:** • Implement conditional statements (if, else, elif) in real-life scenarios • Write programs using loops (for, while, and nested loops) to solve repetitive tasks • Define custom functions, including examples of recursion, use lambda functions for inline operations.	10
III	**Data Structures in Python:** • Perform CRUD operations on lists, tuples, sets, and dictionaries, use list comprehensions to filter and transform data • Perform string manipulations using built-in methods • Introduce Python's collections module with practical examples.	10
IV	**File Handling and Modules:** • Write programs to read, write, and append text files • Work with CSV files using Python's csv module • Read and write JSON files to store structured data • Explore built-in modules like os, math, and datetime, create and import custom modules.	10
V	**Object-Oriented Programming (OOP) in Python:** • Define classes and create objects with attributes and methods • Implement encapsulation, inheritance, and polymorphism • Work with magic methods (e.g., init , str) and operator overloading, • Write programs to handle exceptions using try, except, and finally.	10
VI	**Advanced Python and Applications:** • Use NumPy for numerical operations and Pandas for data analysis • Create basic visualizations using Matplotlib, perform web scraping using requests and Beautiful Soup • Automate tasks using the os and shutil modules • Mini-Project: Develop a Python script to solve a real-world problem (e.g., a data analysis script, a file organizer, or a basic web scraper).	10

REFERENCES:

1.	Yashavant Kanetkar, Let Us Python, 2nd Edition, BPB Publications, India, 2020.
2.	Reema Thareja, Python Programming: Using Problem Solving Approach, 1st Edition, Oxford University Press, India, 2017.
3.	R. Nageswara Rao, Core Python Programming, 1st Edition, Dreamtech Press, India, 2018.
4.	Satish Jain & Shashi Singh, Python Programming for Beginners, 1st Edition, BPB Publications, India, 2019.
5.	A. K. Sinha, Python Programming: A Modular Approach with Graphics, Database, Mobile and Web Applications, 1st Edition, BPB Publications, India, 2021.
6.	Yashavant Kanetkar, Let Us Python, 2nd Edition, BPB Publications, India, 2020.

PR2- WEB TECHNOLOGIES LAB

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="7">Total Marks: 50</td><td colspan="2" rowspan="2">Course Code: CEIPC 307</td></tr>
<tr><td>0</td><td>0</td><td>4</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td colspan="2">Practical Assessment</td></tr>
<tr><td colspan="2">Practical</td><td>: 60 Hrs</td><td>End Exam</td><td>15</td></tr>
<tr><td colspan="3"></td><td>Progressive Assessment</td><td>35</td></tr>
<tr><td colspan="2">Pre-Requisite</td><td>: Nil</td><td colspan="2"></td></tr>
<tr><td colspan="2">Credit</td><td>2</td><td>Category of Course</td><td>: PC</td></tr>
</table>

RATIONALE:

This course provides hands-on experience in full-stack web development, covering both frontend and backend technologies. Students will build dynamic web applications using HTML, CSS, JavaScript, PHP, and MySQL while understanding client-server interactions. By the end of the course, they will be able to develop and deploy a fully functional web service application.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Use LAMP Stack (or XAMPP) for web applications.
- Write simple applications with Technologies like HTML, JavaScript, PHP, CSS.
- Connect to Database and get results.
- Develop a fully functional web service application.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	Coding Server Client Programs	4
II	Developing Web Application using HTML, JavaScript	6
III	Developing Advanced Web Application Programs using CSS	10
IV	Practicing PHP: Basics	10
V	Practicing PHP: Web Application Development	10
VI	Practicing PHP: MySql - tiered Applications	10
VII	Developing a fully functional Web Service Application using all the Technologies learned in this course.	10

REFERENCES:

1.	"Web Technologies--A Computer Science Perspective", Jeffrey C.Jackson,
2.	"Internet & World Wide Web How to Program", Deitel, Deitel, Goldberg, Pearson Education
3.	"Web programming- Building Internet Application", Chris Bales
4.	Web Applications: Concepts and Real World Design, Knuckles

TH3A- PROGRAMMING WITH JAVA

L	T	P		Course Code: CEIPE 301A	
3	0	0			
Total Contact Hours				**Theory Assessment**	
Theory	: 45 Hrs		**Total Marks: 100**	End Term Exam	: 70
				Progressive Assessment	: 30
Pre-Requisite	**: Nil**				
Credit	**: 3**			Category of Course	: PE

RATIONALE:

This course provides a comprehensive introduction to Java programming, covering its history, evolution, and core concepts. Students will learn object-oriented programming (OOP) principles, including classes, inheritance, and interfaces, along with essential topics like exception handling, multithreading, and collections framework.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the history, evolution, and core concepts of Java.
- Use datatypes, variables, arrays, operators, and control statements effectively.
- Implement classes, methods, inheritance, packages, and interfaces.
- Implement exception handling to ensure robust and error-free programs.
- Create and manage multiple threads for concurrent programming

DETAILED COURSE CONTENT:

UNITNO.	TOPIC/SUB-TOPIC	ALLOTED TIME (HRS.)
I.	**Introduction to Java** • History and features of Java • Setting up the Java Development Kit (JDK) • Understanding the Java Virtual Machine (JVM) • Basic Java syntax: data types, variables, operators, control flow statements (if-else, switch, loops) • Input/Output operations (using Scanner and System.out) • Writing a Simple Program • Compiling and executing • Create Packaged class	**6**

II.	Introduction To Object Oriented Programming • Basic concepts of OOPs: objects, classes, data encapsulation, inheritance, Polymorphism etc. • Fundamentals of Class & Object, • New Keyword, reference variable, Members methods of a class, • Access modifiers: public, private, protected, default. • Encapsulation: Hiding data through access modifiers. • Access Control • Static Methods, Static Variables, Static Block • Constructor, Overloading methods, Overloading constructors • Using of final keyword, Finalize method	**9**
III.	**Inheritance** • Basics of Inheritance, members accessibility in inheritance • Types of inheritance: Single, multilevel, hierarchical. • Super keyword. • Method overriding. • Final keyword (for classes, methods, and variables). • Dynamic Method Dispatch, Abstract class, Preventing Overriding, Preventing Inheritance • Interfaces: Defining and implementing interfaces.	**6**
IV.	Interfaces • Purpose of interface • Defining an interface • implementing interfaces • Interface reference variables • Interface with variables • Extending interfaces	**5**
V.	**Arrays and Strings** • Arrays: declaring and initializing arrays, accessing array elements • Multidimensional arrays • String manipulation: creating strings, string methods (length, concat, substring, etc.) • Wrapper classes: Integer, Float, Double, etc.	**6**
VI.	**Exception Handling and File Handling** • Exceptions: types of exceptions, try-catch blocks, finally block • Throwing exceptions • Custom exceptions • File input/output: reading from and writing to files • Using File Input Stream, File Output Stream, Buffered Reader, Buffered Writer	**7**

VII.	**Multithreaded programming** • Basics of threads, • Java threaded model, • defining threads using Runnable interface, • defining threads using Thread superclass, • Multiple threads, Thread Priority values, • Thread Synchronization Using synchronized blocks	**6**
	Total	**45**

REFERENCES:

1.	Programming with Java–E.Balagurusamy, Mc Graw Hill Education, India,2021
2.	Core Java: An Integrated Approach–R. Nageswara Rao, Dream tech Press, India,2018
3.	Java Programming–Poornachandra Sarang,Mc Graw Hill Education, India,2019
4.	Object-Oriented Programming with Java–M.T. Savaliya, BPB Publications, India, 2020
5.	“Advanced Java Programming”–K. Somasundaram, McGraw Hill Education, India, 2017

TH3B- IOT USING RASBERRY PI

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="8">Total Marks: 100</td><td colspan="2" rowspan="2">Course Code: CEIPE 301B</td></tr>
<tr><td>3</td><td>0</td><td>0</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td colspan="2">Theory Assessment</td></tr>
<tr><td colspan="2">Theory</td><td>: 45 Hrs</td><td>End Term Exam</td><td>70</td></tr>
<tr><td colspan="3"></td><td>Progressive Assessment</td><td>30</td></tr>
<tr><td colspan="2" rowspan="2">Pre-Requisite</td><td rowspan="2">: Nil</td><td colspan="2"></td></tr>
<tr><td colspan="2"></td></tr>
<tr><td colspan="2">Credit</td><td>3</td><td>Category of Course</td><td>: PE</td></tr>
</table>

RATIONALE:

The IoT Using Raspberry Pi course provides a theoretical foundation for understanding the role of Raspberry Pi in IoT architectures. It covers the principles of embedded systems, IoT communication protocols, cloud integration, and security. This course enables students to conceptualize IoT applications and analyze their impact on various domains.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the architecture and working principles of Raspberry Pi in IoT systems.
- Describe the role of communication protocols like MQTT, HTTP, and CoAP in IoT.
- Analyze sensor data processing techniques and cloud-based data management.
- Explain the security challenges in IoT and strategies for secure communication.
- Evaluate different IoT applications and their impact on industries.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to IoT and Raspberry Pi:** Introduction to IoT: Concepts, Applications, and Market Trends, Overview of IoT Hardware: Microcontrollers vs. Microprocessors, Introduction to Raspberry Pi: Features, Models, and Setup, Installing Raspberry Pi OS and Basic Configuration, Working with Raspberry Pi GPIO Pins, Basic Linux Commands and File System in Raspberry Pi.	9
II	**Interfacing Sensors and Actuators:** Understanding Digital and Analog Sensors, Interfacing Temperature and Humidity Sensor (DHT11), Interfacing Ultrasonic Sensor for Distance Measurement, Reading Data from Light Sensors (LDR), Working with Actuators: Controlling Motors and Relays.	9
III	**IoT Communication and Data Transmission:** Introduction to IoT Communication Protocols (MQTT, HTTP, WebSockets), Setting Up MQTT Broker on Raspberry Pi, Sending and Receiving MQTT Messages, Sending Sensor Data to the Cloud, Working with HTTP APIs on Raspberry Pi, Posting Data to a Web Server Using HTTP Requests.	9

IV	**Wireless Connectivity and Cloud Integration:** Connecting Raspberry Pi to Wi-Fi Networks, Bluetooth Communication with Raspberry Pi, Using LoRa and Zigbee for IoT Communication, Implementing IoT Cloud Services, Storing Sensor Data in Databases, Data Logging and Real-Time Graphing.	9
V	**Security and IoT Applications:** Introduction to IoT Security, Securing Raspberry Pi Using Firewalls and Encryption, Implementing Secure Communication with HTTPS and TLS, Authentication and Access Control in IoT Systems	9

REFERENCES:

1	Simon Monk, Programming the Raspberry Pi: Getting Started with Python, 3rd Edition, McGraw-Hill, New York, 2021.
2	Donald Norris, The Internet of Things: Do-It-Yourself at Home Projects for Arduino, Raspberry Pi, and BeagleBone, 3rd Edition, McGraw-Hill, New York, 2020.
3	Matt Richardson & Shawn Wallace, Getting Started with Raspberry Pi, 3rd Edition, O'Reilly Media, California, 2019.
4	Peter Waher, Mastering Internet of Things: Design and Create Your Own IoT Applications, 3rd Edition, Packt Publishing, Birmingham, 2022.
5	Charles Bell, IoT: Building Arduino-Based Projects, 3rd Edition, Apress, New York, 2021.

TH3C- INDUSTRY IOT AND IOT STANDARDS

L	T	P		
3	0	0		**Course Code: CEIPE301C**
Total Contact Hours				**Theory Assessment**
Theory : 45 Hrs			**Total Marks: 100**	End Term Exam 70
				Progressive Assessment 30
Pre-Requisite : Nil				
Credit 3				Category of Course : PE

RATIONALE:

Industrial IoT (IIoT) integrates smart sensors, communication protocols, and data analytics to enhance automation and operational efficiency in industries. Understanding IIoT standards and frameworks ensures compliance, security, and interoperability across industrial systems. This course provides a theoretical foundation in IIoT technologies, communication protocols, security measures, and data management strategies.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the fundamental concepts and architecture of Industrial IoT (IIoT).
- Identify various IIoT communication protocols and their industrial applications.
- Analyze IoT and IIoT security challenges, standards, and regulatory frameworks.
- Describe cloud and edge computing strategies for IIoT data management and analytics.
- Examine Industry 4.0 trends, digital twins, and smart factory concepts.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to Industrial IoT (IIoT):** Introduction to IIoT and its significance in Industry 4.0, Differences between IoT and Industrial IoT, IIoT Architecture and Components, Applications of IIoT in Manufacturing, Healthcare, Energy, and Smart Cities, Industrial Automation and IIoT, Challenges and Opportunities in IIoT.	8
II	**Communication Protocols in IIoT:** Overview of Communication Technologies in IIoT, Wired vs. Wireless Communication in IIoT, Basics of Industrial Ethernet, Fieldbus, and Modbus, Wireless Technologies: Wi- Fi, Bluetooth, Zigbee, LoRaWAN, NB-IoT, IIoT-specific Protocols: MQTT, CoAP, OPC-UA, AMQP.	8

III	**IIoT Standards and Security:** IoT and IIoT Standards: ISO, IEEE, IEC, ITU-T, Industrial IoT Frameworks: Industrial Internet Consortium (IIC), EdgeX Foundry, Compliance and Regulatory Requirements, IIoT Security Challenges: Cyber Threats in Industrial Environments, Security Frameworks: NIST, ISA/IEC 62443, Secure Communication in IIoT Networks.	8
IV	**IIoT Data Management and Analytics: Industrial** Data Collection, Processing, and Storage, Cloud vs. Edge Computing in IIoT, Data Preprocessing Techniques for IIoT, Machine Learning and AI in IIoT, Predictive Maintenance using IIoT Analytics.	7
V	**IIoT Hardware and Platforms:** Overview of IIoT Hardware: Sensors, Actuators, Edge Devices, Microcontrollers and Microprocessors in IIoT: Raspberry Pi, ESP32, Industrial PLCs, IIoT Gateways and Edge Computing Devices, Industrial IoT Cloud Platforms, Connecting Devices to IIoT Cloud Platforms	7
VI	**Industry 4.0 and IIoT Use Cases:** Introduction to Industry 4.0 and Smart Factories, Digital Twins and IIoT, Robotics and Automation in IIoT, Future Trends in IIoT and Industry 5.0	7

REFERENCES:

1.	Sabina Jeschke, Christian Brecher, Houbing Song, Danda B. Rawat, Industrial Internet of Things: Cyber manufacturing Systems, 1st Edition, Springer, Cham, 2016.
2.	Rajkumar Buyya, Amir Vahid Dastjerdi, Internet of Things: Principles and Paradigms, 1st Edition, Elsevier, Amsterdam, 2016.
3.	Alasdair Gilchrist, Industry 4.0: The Industrial Internet of Things, 1st Edition, Apress, Berkeley, 2016.
4.	Maciej Kranz, Building the Internet of Things: Implement New Business Models, Disrupt Competitors, Transform Your Industry, 1st Edition, Wiley, New York, 2016.
5.	Olivier Hersent, David Boswarthick, Omar Elloumi, The Internet of Things: Key Applications and Protocols, 1st Edition, Wiley, West Sussex, 2012.

TH4A- DATA SCIENCE AND DATA MANAGEMENT FOR IOT

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="7">Total Marks: 100</td><td colspan="2">Course Code: CEIPE303A</td></tr>
<tr><td>3</td><td>0</td><td>0</td><td colspan="2">Theory Assessment</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td>End Term Exam</td><td>70</td></tr>
<tr><td colspan="2">Theory</td><td>: 45 Hrs</td><td>Progressive Assessment</td><td>30</td></tr>
<tr><td colspan="3"></td><td colspan="2"></td></tr>
<tr><td colspan="2">Pre-Requisite</td><td>: Nil</td><td colspan="2"></td></tr>
<tr><td colspan="2">Credit</td><td>3</td><td>Category of Course</td><td>: PE</td></tr>
</table>

RATIONALE:

The Data Science and Data Management for IoT course provides students with the skills to handle, process, and analyze large-scale IoT-generated data. It focuses on data storage, real- time processing, and machine learning applications in IoT ecosystems. This course prepares students to develop efficient data-driven solutions for smart cities, healthcare, and industrial automation.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the fundamentals of IoT data generation, collection, and management.
- Implement data storage, preprocessing, and cloud-based solutions for IoT applications.
- Apply machine learning and big data analytics to extract insights from IoT data.
- Develop real-time monitoring and visualization dashboards for IoT systems.
- Ensure data security, privacy, and compliance in IoT data management.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to IoT and Data Science:** Overview of IoT and its applications, Basics of Data Science and its role in IoT, Types of IoT data: Structured, Unstructured, and Semi-structured, Challenges in IoT data collection and management	7
II	**Data Collection and Preprocessing in IoT:** IoT sensor data acquisition techniques, Data preprocessing: Cleaning, normalization, and transformation, Real-time data handling vs. batch processing, Introduction to Edge Computing for IoT Data	7
III	**IoT Data Storage and Management:** Overview of IoT Databases: SQL vs. NoSQL, Cloud storage solutions for IoT, Distributed storage systems for IoT data, Security and privacy considerations in IoT data storage	7
IV	**Data Analytics for IoT:** Exploratory Data Analysis (EDA) on IoT datasets, Time-series data analysis techniques, Machine Learning for IoT applications	8

V	**Big Data and IoT:** Introduction to Big Data frameworks: Hadoop, Spark, Real-time processing of IoT data using Apache Kafka and Spark	8
	Streaming, Scalable IoT data architectures	
VI	**Data Visualization and Reporting:** Visualization tools for IoT Data: Power BI, Tableau, Matplotlib, Dashboard creation for real-time IoT data monitoring, Implementing alerts and notifications for IoT systems	8

REFERENCES:

1.	Cathy O'Neil, Rachel Schutt, Doing Data Science, 1st Edition, O'Reilly Media, Sebastopol, CA, 2013.
2.	Jules J. Berman, Principles and Practice of Big Data: Preparing, Sharing, and Analyzing Complex Information, 2nd Edition, Academic Press, Cambridge, MA, 2018.
3.	Pethuru Raj, Anupama C. Raman, The Internet of Things: Enabling Technologies, Platforms, and Use Cases, 1st Edition, CRC Press, Boca Raton, FL, 2017.
4.	Milan Milenkovic, Cloud Computing and Internet of Things: A Roadmap for Smart Environments, 1st Edition, Springer, Cham, Switzerland, 2020.
5.	Benjamin Bengfort, Rebecca Bilbro, Tony Ojeda, Applied Text Analysis with Python: Enabling Language-Aware Data Products with Machine Learning, 1st Edition, O'Reilly Media, Sebastopol, CA, 2018.

TH4B- IOT COMMUNICATION TECHNOLOGY

L	T	P		
3	0	0	**Total Marks: 100**	**Course Code: CEIPE 303B**
Total Contact Hours				**Theory Assessment**
Theory : 45 Hrs				End Term Exam 70
				Progressive Assessment 30
Pre-Requisite : Nil				
Credit 3				Category of Course : PE

RATIONALE:

The IoT Communication Technology course provides an in-depth understanding of various communication protocols and networking technologies essential for IoT systems. It covers wireless communication standards, cloud integration, edge computing, and security mechanisms to ensure reliable and efficient IoT data exchange. This course prepares students to design and implement secure and scalable IoT communication networks for real-world applications.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain key IoT communication technologies, including Wi-Fi, Bluetooth, ZigBee, LoRaWAN, and 5G.
- Implement IoT communication protocols such as MQTT, CoAP, and WebSockets.
- Develop secure IoT networks using encryption techniques.
- Integrate IoT devices with cloud platforms for data processing and analytics.
- Apply IoT communication concepts in real-world scenarios like smart cities, healthcare, and industrial IoT.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to IoT Communication**: Overview of IoT Communication, IoT Communication Requirements & Challenges, Types of IoT Networks: PAN, LAN, WAN, IoT Communication Models & Protocols, OSI Model and TCP/IP in IoT	7
II	**Wireless Communication Technologies for IoT:** Wi-Fi and Bluetooth in IoT, ZigBee, Z-Wave, and LoRaWAN, 5G and Cellular IoT (NB-IoT, LTE-M), RF Communication and Near Field Communication (NFC), Comparative Analysis of Wireless IoT Technologies	7

III	**IoT Networking Protocols:** MQTT (Message Queuing Telemetry Transport), CoAP (Constrained Application Protocol), HTTP and WebSockets for IoT, AMQP (Advanced Message Queuing Protocol), Implementing MQTT and CoAP in IoT Applications	7
IV	**IoT Edge and Cloud Connectivity:** Edge Computing vs. Cloud Computing in IoT, IoT Gateways and Their Role in Communication, IoT Data Transmission to Cloud Platforms, Cloud-Based IoT Services, Real-time Data Streaming and Processing	8
V	**IoT Security and Privacy:** Security Challenges in IoT Communication, Secure IoT Communication Protocols (DTLS, TLS, IPSec), Data Encryption in IoT Networks, Authentication Mechanisms for IoT Devices	8
VI	**IoT Communication in Industrial Applications:** Smart Cities and Smart Homes, Industrial IoT (IIoT) and Smart Manufacturing, IoT in Healthcare and Smart Agriculture, Vehicle-to-Everything (V2X) Communication, IoT Communication in Supply Chain and Logistics	8

REFERENCES:

1.	Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-on Approach, 1st Edition, Universal Press, Georgia, 2014.
2.	Pethuru Raj, Anupama C. Raman, The Internet of Things: Enabling Technologies, Platforms, and Use Cases, 1st Edition, CRC Press, Boca Raton, 2017.
3.	Raj Kamal, Internet of Things: Architecture and Design Principles, 1st Edition, McGraw Hill, New Delhi, 2017.
4.	Olivier Hersent, David Boswarthick, Omar Elloumi, The Internet of Things: Key Applications and Protocols, 2nd Edition, Wiley, West Sussex, 2012.
5.	Adrian McEwen, Hakim Cassimally, Designing the Internet of Things, 1st Edition, Wiley, West Sussex, 2013.

TH4C- DRONE TECHNOLOGY AND WSN

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="8">Total Marks: 100</td><td colspan="2" rowspan="2">Course Code: CEIPE 303C</td></tr>
<tr><td>3</td><td>0</td><td>0</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td colspan="2">Theory Assessment</td></tr>
<tr><td colspan="2">Theory</td><td>: 45 Hrs</td><td>End Term Exam</td><td>70</td></tr>
<tr><td colspan="3"></td><td>Progressive Assessment</td><td>30</td></tr>
<tr><td colspan="2" rowspan="2">Pre-Requisite</td><td rowspan="2">: Nil</td><td colspan="2"></td></tr>
<tr><td colspan="2"></td></tr>
<tr><td colspan="2">Credit</td><td>3</td><td>Category of Course</td><td>: PE</td></tr>
</table>

RATIONALE:

The Drone Technology and Wireless Sensor Networks (WSN) course provides students with a fundamental understanding of UAV systems, drone hardware, flight control, and wireless communication technologies. It equips learners with practical knowledge of integrating WSN with drones for real-time data acquisition, surveillance, and automation. The course emphasizes hands-on experience in drone assembly, programming, and IoT applications for smart environments.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the fundamental concepts of drone technology, flight control, and navigation systems.
- Develop practical skills in assembling and programming drones for autonomous operations.
- Describe Wireless Sensor Networks (WSN) and their integration with drones for real-time monitoring.
- Implement drone-based applications in agriculture, defense, and smart city environments.
- Develop security and communication strategies for efficient UAV-WSN interaction.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to Drone Technology:** Basics of UAVs (Unmanned Aerial Vehicles) and their classifications, History and evolution of drones, Applications of drones in agriculture, defense, logistics, and surveillance, Key components of drones: Frame, motors, propellers, flight controllers, ESC, and batteries, Drone communication protocols	7
II	**Drone Hardware and Assembly:** Understanding drone components and assembling a basic quadcopter, Power distribution and motor-ESC connections, Flight controllers: Pixhawk, KK2, and Open Pilot, Calibrating sensors: IMU, GPS, barometer, and magnetometer, Safety measures and troubleshooting	7

III	**Flight Control and Programming:** Introduction to flight control systems, Drone stabilization and GPS navigation, Autonomous flight programming using ArduPilot/Mission Planner, Communication protocols: PWM, SBUS, and MAVLink, Simulating drone flights using Gazebo or AirSim	7
IV	**Wireless Sensor Networks (WSN) Basics:** Introduction to Wireless Sensor Networks (WSN), Sensor types and their roles in IoT and drones, WSN architecture: Nodes, gateways, and cloud connectivity, Communication protocols in WSN: ZigBee, LoRa, Bluetooth, and Wi-Fi, Energy-efficient sensor networks	8
V	**Interfacing WSN with Drones:** Integrating sensors with drones for data collection, Real-time data acquisition and transmission, Drone-based environmental monitoring using WSN, Case study: Precision agriculture and disaster management	8
VI	**Advanced Applications and Security:** AI and Machine Learning applications in drone navigation, Swarm intelligence and cooperative drones, Cybersecurity challenges in drone communication	8

REFERENCES:

1.	Paul G. Fahlstrom, Thomas J. Gleason, Introduction to UAV Systems, 4th Edition, Wiley, Hoboken, NJ, 2019.
2.	Reg Austin, Unmanned Aircraft Systems: UAVS Design, Development, and Deployment, 2nd Edition, Wiley, West Sussex, UK, 2011.
3.	Sudip Misra, Anandarup Mukherjee, Arijit Roy, Introduction to Wireless Sensor Networks, 1st Edition, Cambridge University Press, Cambridge, UK, 2021.
4.	K Raghunandan, V. Udayashankara, Wireless Sensor Networks: APractical Approach, 1st Edition, McGraw-Hill, New Delhi, India, 2018.

PR3A- PROGRAMMING WITH JAVA LAB

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="8">Total Marks: 50</td><td colspan="2" rowspan="2">Course Code: CEIPE305A</td></tr>
<tr><td>0</td><td>0</td><td>4</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td colspan="2">Practical Assessment</td></tr>
<tr><td colspan="2">Practical</td><td>: 60 Hrs.</td><td>End Exam</td><td>: 15</td></tr>
<tr><td colspan="3"></td><td>Progressive Assessment</td><td>: 35</td></tr>
<tr><td colspan="2" rowspan="2">Pre-Requisite</td><td rowspan="2">: Nil</td><td colspan="2"></td></tr>
<tr><td colspan="2"></td></tr>
<tr><td colspan="2">Credit</td><td>: 2</td><td>Category of Course</td><td>: PE</td></tr>
</table>

RATIONALE:

This lab series is designed to provide hands-on experience aligned with core Java programming concepts. Each exercise reinforces theoretical understanding through practical implementation. The structured progression ensures learners build a strong foundation from basics to advanced topics like file input output and frameworks.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the core concepts and syntax of Java programming, including data types, control structures, and operators.
- Apply object-oriented programming principles such as encapsulation, inheritance to create java applications.
- Apply object-oriented programming principles like polymorphism, and abstraction to build modular and reusable code.
- Implement exception handling and multithreading techniques to create robust and concurrent Java applications.
- Implement/utilize Java’s standard libraries and frameworks for effective input/output operations, data manipulation, and utility functions.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Setting Up Java and Writing First Program:** Install Java JDK and configure environment variables. Write, compile, and run a basic “Hello World” program. Demonstrate the use of variables, arrays, and different data types.	6
II	**Expressions and Operators:** Create a program using arithmetic, relational, and logical operators. Practice operator precedence and mixed-type expressions.	6
III	**Control Flow & Looping:** Write programs using if-else, switch, while, do-while, and for loops. Implement examples with break, continue, and nested loops.	6

IV	**Classes, Objects & Constructors:** Define a class with fields and methods. Use constructors (default, parameterized) and method overloading. Demonstrate the use of this and new keyword.	6
V	**Static, Final and Access Control:** Implement static variables, static methods, and static blocks. Use access modifiers (public, private, Protected) and final keyword.	6
VI	**Inheritance and Method Overriding:** Create a class hierarchy demonstrating multilevel inheritance. Use super keyword and show constructor execution sequence. Override methods and implement dynamic method dispatch.	6
VII	**Abstract Classes and Preventing Overriding:** Create an abstract class and implement its abstract methods in a subclass. Use final to prevent Method overriding or class inheritance. Create a packaged class.	6
VIII	**Handling Exceptions:** Write programs using try-catch-finally blocks. Create user-defined exceptions. Demonstrate throw, throws, and nested exception handling.	6
IX	**Working with Interfaces:** Define and implement interfaces. Use interface reference variables and interface inheritance.	6
X	**Using Java Standard Libraries:** Use String, StringBuilder, Math, and Date classes. Practice with Java Collections (Array List, HashMap, etc.). Perform file input/output operations.	6

REFERENCES:

1.	Programming with Java – E. Balagurusamy, McGraw Hill Education, India, 2021
2.	Core Java: An Integrated Approach – R. Nageswara Rao, Dream tech Press, India, 2018
3.	Java Programming– Poornachandra Sarang, McGraw Hill Education, India, 2019
4.	Object-Oriented Programming with Java– M.T. Savaliya, BPB Publications, India, 2020
5.	"Advanced Java Programming" – K. Somasundaram, McGraw Hill Education, India, 2017

PR3B- IOT USING RASBERRY PI LAB

L	T	P		**Course Code: CEIPE 305B**	
0	0	4			
Total Contact Hours				**Practical Assessment**	
Practical	: 60 Hrs		**Total Marks: 50**	End Exam	15
				Progressive Assessment	35
Pre-Requisite	**: Nil**				
Credit	**2**			Category of Course	: PE

RATIONALE:

This lab series offers practical exposure to IoT concepts using Raspberry Pi, bridging theory with real-world applications. Each exercise is designed to develop hands-on skills in sensor interfacing, communication protocols, and data transmission. The labs also emphasize secure IoT system design and cloud integration to build end-to-end smart solutions.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the fundamental concepts and applications of IoT and differentiate between microcontrollers and microprocessors.
- Configure Raspberry Pi, and interact with its GPIO pins using basic Linux commands and Python programming.
- Interface various sensors and actuators with Raspberry Pi to collect, process, and act upon real-time data.
- Develop programs to interface and control a variety of sensors and actuators, enabling the Raspberry Pi to interact with its physical environment effectively.
- Analyze real-time sensor data and perform basic data logging and visualization using Python and built-in Raspberry Pi functionalities.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Getting Started with Raspberry Pi:** Set up Raspberry Pi with Raspberry Pi OS. Explore basic Linux commands and navigate the file system. Configure network settings and test connectivity.	10
II	**GPIO Programming Basics:** Write a Python program to blink an LED using GPIO pins. Understand pin numbering, input/output modes, and basic circuit setup.	10
III	**Interfacing DHT11 Sensor:** Connect and read temperature and humidity data from a DHT11 sensor. Display the sensor data on the terminal or save it to a text file.	10

IV	**Distance Measurement with Ultrasonic Sensor:** Connect an ultrasonic sensor and measure distance using GPIO. Display real-time distance data in the terminal.	10
V	**Light Sensing with LDR:** Interface an LDR with Raspberry Pi via an ADC (e.g., MCP3008). Record light intensity and visualize changes over time.	10
VI	**Controlling Actuators (Motors and Relays):** Control a DC motor and a relay using GPIO pins. Use external power sources and implement safety mechanisms.	10

REFERENCES:

1.	Simon Monk, Programming the Raspberry Pi: Getting Started with Python, 3rd Edition, McGraw-Hill, New York, 2021.
2.	Donald Norris, The Internet of Things: Do-It-Yourself at Home Projects for Arduino, Raspberry Pi, and BeagleBone, 3rd Edition, McGraw-Hill, New York, 2020.
3.	Matt Richardson & Shawn Wallace, Getting Started with Raspberry Pi, 3rd Edition, O'Reilly Media, California, 2019.
4.	Peter Waher, Mastering Internet of Things: Design and Create Your Own IoT Applications, 3rd Edition, Packt Publishing, Birmingham, 2022.
5.	Charles Bell, IoT: Building Arduino-Based Projects, 3rd Edition, Apress, New York, 2021.

PR3C- INDUSTRY IOT AND IOT STANDARDS LAB

L	T	P		Course Code: CEIPE305C	
0	0	4			
Total Contact Hours				**Practical Assessment**	
Practical	: 60 Hrs		**Total Marks: 50**	End Exam	15
				Progressive Assessment	35
Pre-Requisite	**: Nil**				
Credit	**2**			Category of Course	: PE

RATIONALE:

This lab series provides hands-on experience with Industrial IoT concepts, bridging theoretical knowledge with real-world industrial applications. It covers key areas such as sensor interfacing, communication protocols, data handling, and cloud integration. The labs are designed to prepare students for Industry 4.0 by equipping them with practical skills in IIoT system development and deployment.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Explain the architecture and components of Industrial IoT systems, and differentiate IIoT from traditional IoT through real-world industrial applications.
- Evaluate various IIoT communication protocols and demonstrate their use and relevance using industry-standard simulation tools.
- Implement both wired and wireless communication setups between IIoT devices using platforms like Raspberry Pi or ESP32.
- Operate an MQTT-based messaging system for publishing and subscribing to sensor data in an IIoT environment.
- Analyze security challenges in IIoT systems and implement secure communication practices using encryption, authentication, and compliance frameworks.

DETAILED COURSE CONTENTS:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Exploring IIoT Concepts and Architectures:** Create a conceptual diagram of an IIoT system. Compare IoT vs. IIoT using real-world industrial examples. Identify key components in an IIoT architecture (sensors, gateways, cloud, etc.).	10
II	**IIoT Communication Technologies Overview:** Analyze different communication protocols (Modbus, OPC-UA, MQTT) using simulation tools like Node-RED or Wireshark. Create a comparison table showing their use cases and industrial applicability.	10

III	**Wired and Wireless Communication Setup:** Demonstrate communication between two IIoT nodes using UART (wired) and Bluetooth (wireless) using Raspberry Pi or ESP32. Monitor data transmission using serial communication.	10
IV	**Setting Up MQTT for IIoT:** Install Mosquitto MQTT broker. Publish sensor data and subscribe to topics using Python on Raspberry Pi/ESP32.	10
V	**IIoT Security Risk Simulation:** Identify security vulnerabilities in a simulated IIoT environment. Create a checklist of IIoT security best practices based on ISA/IEC 62443 and NIST guidelines.	10
VI	**Simulating Secure Communication:** Set up TLS-enabled MQTT communication using certificates. Validate secure data transmission using packet sniffers.	10

REFERENCES:

1.	Sabina Jeschke, Christian Brecher, Houbing Song, Danda B. Rawat, Industrial Internet of Things: Cybermanufacturing Systems, 1st Edition, Springer, Cham, 2016.
2.	Rajkumar Buyya, Amir Vahid Dastjerdi, Internet of Things: Principles and Paradigms, 1st Edition, Elsevier, Amsterdam, 2016.
3.	Alasdair Gilchrist, Industry 4.0: The Industrial Internet of Things, 1st Edition, Apress, Berkeley, 2016.
4.	Maciej Kranz, Building the Internet of Things: Implement New Business Models, Disrupt Competitors, Transform Your Industry, 1st Edition, Wiley, New York, 2016.
5.	Olivier Hersent, David Boswarthick, Omar Elloumi, The Internet of Things: Key Applications and Protocols, 1st Edition, Wiley, West Sussex, 2012.

TH5A- UNIVERSAL HUMAN VALUES

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="8">Total Marks: 100</td><td rowspan="2">Course Code: OE 301A</td></tr>
<tr><td>3</td><td>0</td><td>0</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td>Theory Assessment</td></tr>
<tr><td colspan="3">Theory : 45Hrs</td><td>End Term Exam 70</td></tr>
<tr><td colspan="3"></td><td>Progressive Assessment 30</td></tr>
<tr><td colspan="3" rowspan="2">Pre-Requisite : Nil</td><td></td></tr>
<tr><td></td></tr>
<tr><td colspan="3">Credit 3</td><td>Category of Course: OE</td></tr>
</table>

RATIONALE:

The Universal Human Values (UHV) course aims to help diploma students develop a strong ethical foundation, nurturing responsible individuals who contribute positively to society. In an era driven by rapid technological advancements, it is crucial for students not only to gain technical expertise but also to cultivate values that promote harmony, respect, and sustainability.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Identify fundamental human aspirations such as happiness and prosperity.
- Differentiate between the self and the body and understand their respective needs.
- Practice self-reflection to improve decision-making, emotional balance, and personal growth.
- Develop respectful and trustworthy relationships within family, friends, and society.
- Explain the role of values like trust, respect, and love in building strong social bonds.
- Promote cooperation and harmony within communities through ethical practices.

DETAILED COURSE CONTENT:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Introduction to Value Education and Human Values -** Concept and Need for Value Education - Understanding the importance of value education in personal and professional life, Differentiating between values and skills. Basic Human Aspirations - Exploring fundamental human aspirations: happiness and prosperity, Methods to achieve these aspirations through right understanding and relationships.	8
II	**Harmony in the Human Being -** Understanding the Self - Differentiating between the 'Self' (I) and the Body, Understanding the needs of the Self and the Body, Harmony of the Self with the Body - Ensuring the harmony of 'I' with the Body, Practices for mental and physical well-being.	8
III	**Harmony in the Family and Society -** Family as the Basic Unit of Society - Understanding values in human relationships, Trust and respect as the foundational values in relationships, Harmony in Society - The concept of an undivided society, Universal human order and world family.	8

IV	**Harmony in Nature and Existence -** Interconnectedness in Nature -Understanding the four orders of nature: material, plant, animal, and human, Mutual fulfillment among these orders, Co-existence in Existence - Holistic perception of harmony in existence, Role of human beings in maintaining environmental balance.	8
V	**Professional Ethics -** Ethical Human Conduct - Integrating values into professional life, Concept of professional ethics and accountability, Case Studies in Professional Ethics - Analyzing real-life scenarios to understand ethical dilemmas, Developing solutions based on universal human values.	8
VI	**Personal Development and Social Responsibility -** Self-Reflection and Self-Exploration - Techniques for self-assessment and personal growth, Setting personal goals aligned with universal values, Social Responsibility - Understanding one's role in society, Participating in community service and social initiatives.	5

REFERENCES:

1.	R. R. Gaur, R. Asthana, G. P. Bagaria, A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019.
2.	R. R. Gaur, R. Asthana, G. P. Bagaria, Teachers' Manual for A Foundation Course in Human Values and Professional Ethics, 2nd Revised Edition, Excel Books, New Delhi, 2019.
3.	A. Nagraj, JeevanVidya: EkParichaya, Amarkantak, 1999.
4.	A. N. Tripathi, Human Values, New Age Intl. Publishers, New Delhi, 2004.
5.	Moral Thinking: An Introduction To Values And Ethics, Vineet Sahu, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc23_hs89/preview

TH5B- LEADERSHIP AND MANAGEMENT SKILLS

L	T	P		
3	0	0	Total Marks: 100	Course Code: OE301B
Total Contact Hours				**Theory Assessment**
Theory : 45Hrs				End Term Exam 70
				Progressive Assessment 30
Pre-Requisite : Nil				
Credit 3				Category of Course: OE

RATIONALE:

This course/subject on Leadership and Management Skills for students undergoing Diploma programmes is an exploration in leading and managing people, majorly in education based on sound and acceptable principles and theories for effective leadership. The leadership skills will enable them to take initiative, guide team efforts, motivate peers, and ensure effective collaboration. They'll learn how to delegate tasks, resolve conflicts, and foster a positive team environment. The management skills will help them in organizing tasks, setting timelines, and ensuring efficient workflow within a team.
It is expected that the students will be able to handle projects with better project outcomes and a earn a more productive learning experience. This will benefit their academic journey, future careers, and overall professional development:

LEARNING OUTCOMES:

After completion of the course, the students will be able to

- Explain the principles of management
- Collaborate across cultures for effective team work
- Communicate with people for a positive work culture
- Demonstrate personal dispositions, skills & abilities of a leader
- Undertake the process of change management
- Design training for staff development
- Adapt suitable leadership style for improved work efficiency.

DETAILED COURSE CONTENT:

Unit No.	Topic/Sub-Topic	Allotted Time (Hours)
I	**Leadership & Management, concept, principles.** • Definition of leadership, management • Leadership theories • Leadership characteristics • Principles of management • Managerial functions • Leader v/s Manager, Leader/Manager traits and character • Leadership Styles	**10**

II	**Human Resource Management in Organizations** • Human Resource Management: Meaning, Nature, Objectives, Scope • Job & Job analysis. • Staff Development: Need and Objectives of Staff Development, Approaches • Training & development • Organizational Development: Components of OD process. • Learning organization	10
III	**Personal disposition, skills & abilities of leaders** • Self-awareness • Leadership characteristics, traits • Leadership skills & abilities • Emotional intelligence & its components, importance in leadership • Communication skills for effective leadership, barriers to effective communication, Active Listening, Mindful listening. • Leading & Mentorship – Influencing & mentoring	09
IV	**Leader's role in Motivating, Inspiring and Transformative leadership, nurturing team-work** • Goal setting & leadership • Transformative Leadership, vision & envisioning • Motivational role of leader in people management • Group & team • Team dynamics • Conflict management, strategies in managing conflicts	08
V	**Change Management & Leadership** • Models of change • Forces driving change • Change Management – process, goal, importance • The process of change happening in an organization • Key aspects of leadership in change management – responsibilities of a change leader.	08

SUGGESTED ACTIVITIES:

- Group/individual presentation on the basic principles of leadership and management, Discussion on readings - Individual or group presentation of assigned topics in class on leadership and management principles and theories.
- Activities on Envisioning, Goal setting
- ACTION PLAN to be prepared

REFERENCES:

1.	Theories of Educational Leadership and Management (3rd ed.), by Bush, Tony (2003). SAGE Publications, Ltd.
2.	The inspiring leader: unlocking the secrets of how extraordinary leaders motivate. By Zenger, John, Joseph Folkman, and Scott Edinger (2009). New York: McGraw Hill Press.
3.	Knowing yourself. On becoming a leader: the leadership classic. By Bennis, Warren (2009). New York: Basic Books.
4.	Leading Change. By P. Kotter, Harvard Business, 2012.
5.	The Fifth Discipline. By Peter M. Senge, Crwon Currency, 2006.
6.	The Leadership Sutra: An Indian Approach to Power. By Devdutt Pattanaik, – Penguin Random House, 2017.
7.	Leadership and Management. By Dr. A. Chandra Mohan. Himalaya Publishing House, 2010.

TH5C- PROFESSIONAL SKILLS

L	T	P		
3	0	0	Total Marks: 100	**Course Code: OE301C**
Total Contact Hours				**Theory Assessment**
Theory : 45Hrs				End Term Exam 70
				Progressive Assessment 30
Pre-Requisite : Nil				
Credit 3				Category of Course: OE

RATIONALE:

The term, "Professional skills" carries significant weight in the job market and career development. This open elective course explores various types of professional skills, their significance, and how they can be cultivated and harnessed for career progression. By understanding the landscape of professional skills, student can better position himself or herself for success in the competitive job market. It is crucial to continuously update and adapt the professional skills to stay ahead in a rapidly changing work environment. By investing in professional development, one can enhance employability and open doors to new opportunities.

LEARNING OUTCOMES:

After completion of the course, the students will be able to

- Demonstrate Self-competency and Confidence
- Practice Emotional Competency
- Work in a team work or in collaboration
- Demonstrate problem solving and decision making skills
- Apply time management strategies and techniques effectively
- Apply professional ethics and integrity in professional and personal life

UNIT NO.	CONTENT	ALLOTTED TIME (HOURS)
I Communication Skills:	• Active listening • Verbal and non-verbal communication • Written communication • Presentation skills • Conflict resolution	08
II Teamwork and Collaboration:	• Building trust within a team • Effective collaboration strategies • Role delegation and responsibility sharing • Conflict resolution within a team	08
III Problem-Solving:	• Identifying root causes of issues • Generating solutions and evaluating options	08

	• Decision-making under pressure • Critical thinking skills • Triple constraint issues	
IV Time Management:	• Prioritization and task management • Setting realistic deadlines • Effective time planning and organization	06
V Emotional Intelligence:	• Self-awareness and emotional regulation • Empathy and understanding others' emotions • Managing interpersonal relationships • Motivation • Social skills • Emotional Intelligence (EQ) • Stress management	08
VI Professional Ethics and Integrity:	• Workplace ethics and code of conduct • Confidentiality and data privacy • Professional accountability- • Important Considerations:	07

REFERENCES:

1. Dr. Vitthal Gore: Professional Skills for 21st Century: A Key to Success: Blue Rose-ACADEMIC
2. The ACE of Soft Skills: Attitude, Communication and Etiquette for Success: PEARSON
3. The essence of Leadership: S. Manikutty: Bloomsbury

SUMMER INTERNSHIP - II

<table>
<tr><td>L</td><td>T</td><td>P</td><td rowspan="8">Total Marks: 50</td><td colspan="2" rowspan="2">Course Code: SI 301</td></tr>
<tr><td>0</td><td>0</td><td>0</td></tr>
<tr><td colspan="3">Total Contact Hours</td><td colspan="2">Internship Assessment</td></tr>
<tr><td colspan="2">Internship</td><td>: 3-4 weeks</td><td>End Term Exam</td><td>15</td></tr>
<tr><td colspan="3"></td><td>Progressive Assessment</td><td>35</td></tr>
<tr><td colspan="2" rowspan="2">Pre-Requisite</td><td rowspan="2">: Nil</td><td colspan="2"></td></tr>
<tr><td colspan="2"></td></tr>
<tr><td colspan="2">Credit</td><td>2</td><td>Category of Course</td><td>: SI</td></tr>
</table>

Duration: 3-4 weeks after 4th Semester

RATIONALE:

The Summer Internship course includes activities for promoting industrial interaction at the diploma level in technical institutes. The main aim is to enhance employability skills of the students passing out from Technical Institutions. It allows students to gain direct experience, which cannot be simulated in the classroom environment. Hence, this course aims to create competent professionals for the industry. The internship experience will augment outcome-based learning process and inculcate various attributes in a student in line with the graduate attributes defined by the NBA. Students can also use internships to test their interest in and aptitude for different career areas. This enables students to find out where they might fit best in a professional environment.

LEARNING OUTCOMES:

After completion of the course, the students will be able to:

- Integrate theory and practice of current technological developments relevant to the subject area.
- Assess interests and abilities in their field of study for exploring career alternatives.
- Develop work habits and attitudes necessary for job success.
- Develop communication, interpersonal and other critical skills in the job interview process.
- Identify social, economic and administrative considerations that influence the working environment of industrial organizations.

DETAILED COURSE CONTENTS SUGGESTED

ACTIVITIES:

Activity Heads: Industrial/Govt./ NGO/MSME/ Rural Internship/ Innovation / Entrepreneurship

During the summer vacation after 4th semester, students are ready for industrial experience. They may choose to undergo Internship / Innovation / Entrepreneurship related activities. Students may either choose to work on innovation or entrepreneurial activities resulting in start-up or undergo internship with industry/ NGO's/ Government organizations/ Micro/ Small/ Medium enterprises to make themselves ready for the industry. In case student want to pursue their family business and do not want to undergo internship, a declaration by the parents to be submitted directly to the TPO.

As per guidelines of AICTE, assessment to be done based on the following major activity heads:

I. Innovation / IPR / Entrepreneurship • Participation in innovation related completions for eg. Hackathons etc. to be evaluated by Faculty Mentor. • Development of new product/ Business Plan/ registration of start-up to be evaluated by Program Head. • Participation in all the activities of Institute's Innovation Council (ICC) for eg: IPR workshop/ Leadership Talks/ Idea/ Design/ Innovation/ Business Completion/ Technical Expos etc., evaluated by President/ Convener of ICC. • Work experience at family business – to be evaluated by TPO.
II. Internship • Internship with Industry/ Govt. / NGO/ PSU/ Any Micro/ Small/ Medium enterprise/ Online Internship to be evaluated by Faculty Mentor/ TPO/ Industry supervisor.

PR4- MAJOR PROJECT

L	T	P		
0	0	4		Course Code: PR 301
Total Contact Hours			**Total Marks: 50**	**Project Assessment**
Project : 60 Hrs				End Exam : 15
				Progressive Assessment : 35
Pre-Requisite : Nil				
Credit : 2				Category of Course : Project

RATIONALE:

The project work is a part of teaching learning process. In this course, the role of teachers is a facilitator or coordinator and the students will select the topic, perform design work, place the indent, procure or purchase the material either from departmental store or from the local market. The leadership quality, co-ordination of job and maintaining good communal harmony is an important factor of this type of activity. Project Work is a learning experience, which aims to provide students with the opportunity to synthesize knowledge from various areas of learning, and critically and creatively apply it to real life situations. This process, which enhances student's knowledge and enables them to acquire skills like collaboration, communication and independent learning, prepares them for lifelong learning and the challenges ahead. The success of the project is not the ultimate goal. The group, who is successful in obtaining the good output, should definitely be credited but they must be evaluated for the other components of the activity.

LEARNING OUTCOMES:

After the completion of this course, the students will be able to:

- Select suitable topic, problem and plan of action
- Apply theory and practices to investigate and solve industry / society related problems
- Maintain good relation among the peer groups
- Demonstrate leadership quality
- Estimate cost involvement
- Develop plan for effective utilization of time

GUIDELINES FOR MAJOR PROJECT

• The project work may involve the designing a model or upgrading an existing system. The design is to be implemented into a working model.
• A project work may be carried out by a team of 3 to 5 students with a well-defined role of each student within the team. The group will select a project with approval of team of teachers & the guide.
• Each group must prepare project proposal that includes project title, group members, sponsor details (if any), detailed problem definition, area, abstract, details of existing similar systems if any, scope of the project and software-hardware requirements.
• Major projects should be based on real/ live problems of the Industry/ Govt./ NGO/ MSME/Rural Sector or an innovative idea having the potential of a Start-up. Main objective of the major project is to provide the students with an opportunity to develop a complete project.

- The major project is distributed in two consecutive semesters; students can get ample time to realize a complete project with documentation or transform their ideas of start-ups into reality. The requirement analysis and designing part of the project may be completed in the 5th semester.

- A project report including all necessary documents such as Requirement Analysis, Design specifications, Project Plan, Design Modelling, test plan etc. need to be prepared and submitted for progressive assessment.

- For assessment, following components may be considered:
    - Synopsis and Project Title selection
    - Initiative in Performing Project tasks
    - Sense of responsibility and punctuality
    - Outcome of the completed stages of the Project
    - Communication and presentation skills
    - Interpersonal skills
    - Report writing skills
    - Viva voce

Organization of Project Report

1. Cover page:
   It should contain the following (in order)
    i. Title of the Project
    ii. Submitted in partial fulfilment of the requirements for the Diploma in <Branch Name>
    iii. By Name of the Student(s)
    iv. Logo of the Institution
    v. Branch Name/Depart Name and Institution Name with Address
    vi. Academic Year
2. 1st Inner page

Certificate:
It should contain he following
“This is to certify that the work in this Project Report entitled <Project Title> by <Name of the Student (s)> has/have been carried out under my supervision in partial fulfilment of the requirements for the Diploma in <Branch Name> during session <session> in <Branch/Department Name> of <Institute Name> and this work is the original work of the above student(s).”

Seal and signature of the Supervisor/Guide with date

3. 2nd Inner Page Declaration by the Student(s)

“I declare that this project entitled <Name of the Project> is my own work, except were indicated through the proper use of citations and references, and has not been submitted in any form for another degree or diploma at any other institute. Information derived from the published and unpublished work of others has been acknowledged in the text and a list of references is given.”

Signature of the Student with date

4. 3rd Inner Page Acknowledgement by the Student(s)
5. Contents.
6. Chapter wise arrangement of Reports
7. Last Chapter: Conclusion

It should contain
  i. Conclusion
  ii. Limitations
  iii. Scope for further Improvement

8. References